Current Clarity Stage 12: Water Reflectance Engine

Diagnostic only.

- Production Current Clarity is unchanged, and nothing was deployed.
- No Sector field observation was used, and the P2 fresh holdout was not touched.
- Water Temperature was not started.
- Michael's Guntersville survey stays an independent acceptance test.

Erratum (reissued 2026-09-30).

- **The bug.** In the first issue, the clear-water row and the CLEAR WATER ERROR line reported the in-situ "clear" water type (low turbidity and chlorophyll, 268 rows). A key collision in `models12.py` had overwritten the Secchi class clear ≥ 6 ft.
- **The corrected Secchi-clear figures** (574 rows at 412 lakes): ×1.56 and −3.3 ft for the engine, against ×1.97 and −5.2 ft for production.
- **Nothing else changed:** the engine, every fit and every other row are the same.

Freeze. The pre-registration was hashed at **2026-09-30T14:45:17Z** (sha256 `474a8cd1…`), before any Stage 12 value existed. The certification protocol was hashed at 15:04:45Z (`1e8040b9…`), before any certification site or visit file was fetched. Five addenda, A–D and B2, were each hashed before the data they govern was examined (section 2).

The answer

The new reflectance improves the clarity estimate wherever it can read the water. It reads only three stations in four, and it does not fix the red band.

- **The configuration.** ACOLITE with the FFSS sky-dome interface correction (K4), surface pressure from each lake's elevation, and S2C harmonised at the water-reflectance level.
- **The gain.** C3 refitted on this reflectance is the best model, and no simpler one is equivalent: held-out error **×1.47**, against **×1.66** for today's production on the same 1,206 rows at 792 lakes.
  - **The reflectance itself buys most of it.** The same C3 on Sen2Cor scores ×1.58 on those rows (difference −0.031, CI −0.045 to −0.016).
  - **Clear water is where it shows:** ×1.56 against ×1.97, and a median error of −3.3 ft against −5.2 ft for production. Routing adds nothing (0.8%).
- **What it does not fix:**
  - **Coverage.** ACOLITE gives a valid disk at only **76%** of stations. Sen2Cor has one at all of them.
  - **The red band.** No interface option cut the red bias enough (K4 +0.22 against +0.25). The two that cut it to +0.15 lost too many retrievals.
  - **Turbidity.** It is not better overall: better in clear water, worse in murky water.
  - **Sentinel-2C.** It is only partly harmonised, and the engine's class still flips on 11% of identical S2A/S2C pixels.
- **Capability, historical and strict:**
  - exact feet **0/10**;
  - clarity class **1/10** (clear and sediment PARTIAL).
  - **Stained and limited water are the weakest classes on every input tested** (recall 0.2–0.3).
- **The blind certification set** was built without reading a single Secchi value: 289 observations, 272 lakes, 39 states. It is **not ready**: short of 400 observations, of stained, sediment and limited water, and of the cool season.

1. What was done

Interface experiment	5 frozen ACOLITE configurations against **46 AERONET-OC matchups** at 5 sites, with a 60-lake inland-turbidity guard (Guntersville excluded)
Harmonisation	Derived from the 12 Stage 11 Sentinel-2A/2C tandem pairs only, and validated leave-one-pair-out on both paths (Sen2Cor production and ACOLITE)
Corpus	**1,631 station-days** (1,015 lakes) through the new reflectance: 1,343 L1C scenes and 1,504 crops, of which 1,478 ran and 100 used the angle-grid fallback. Guntersville is kept apart (45 rows).
Models	9 candidates, fitted on training lake-folds only (5 whole-lake folds), with Guntersville excluded from every fit and selection
Capabilities	Exact feet and clarity class, scored separately on historical lake-held-out rows
Certification	A blind external protocol and sample, built without reading any Secchi value
Not pursued	C2RCC, C2X, P1-NIR, routing for its own sake, Landsat and OLCI fusion, temporal replacement, and public depth. No Stage 12 evidence reopened them.

2. Deviations and addenda

- **A (harmonisation, 15:04Z).** A technical check on one pair showed that TOA harmonisation does not carry through ACOLITE's dark-spectrum aerosol step. The addendum added a water-reflectance-level method (V-ρw). It also made the ACOLITE-path verdict use the method that would be adopted operationally, not the V-full emulation.
- **B and B2 (certification matching, 15:31Z and 16:32Z).**
  - Earth Search's production collection has **no scenes at all for 2017 or 2022**, which are the two NLA survey years.
  - Matching for those years uses Earth Search's older `sentinel-2-l2a` collection.
  - B's first reading of that collection's offset flag was wrong: it double-subtracted 0.1 from 2022. B2 corrected it before any match was counted.
- **C (surface pressure, 15:59Z).**
  - At a 2,318 m lake, every pixel was flagged negative, because ACOLITE had used sea-level pressure.
  - Every corpus crop now sets its elevation from USGS 3DEP. ACOLITE converts it to pressure.
  - The 59 rows already processed were discarded.
  - AERONET sites sit at 0–174 m, so Part 1 is unaffected. At Lake Erie, B04 changes by +2% and the pooled error by +0.013.
- **D (viewing geometry, 17:23Z).** Older L1C baselines keep detector footprints as GML, which the GDAL stand-in cannot read. Those crops fall back to ACOLITE's angle-grid geometry.
- **Smaller notes.**
  - The NLA 2022 site file writes the sampled flag as `YES`, not `Y`.
  - SYKE was queried in bulk rather than lake by lake, under the same rules.
  - A candidate that "fails to run" means an ACOLITE error or an all-NaN output. Losing valid station pixels is judged by criterion 3, not as a failure to run.

3. Part 1: ACOLITE's air–water interface options

	Setting	Valid	B02	B03	B04	B05	B02–B05	Guard |e_T|	Verdict
K0	Stage 11 (no interface term)	96%	0.271 (+0.271)	0.174 (+0.174)	0.254 (+0.254)	0.109 (+0.109)	**0.210**	0.279	baseline
K1	interface: default (RSKY LUT, wind 2 m/s)	87%	0.208 (+0.208)	0.121 (+0.118)	0.152 (+0.152)	0.116 (-0.033)	**0.148**	0.305	not eligible: valid rate
K2	interface: 6SV rough surface	89%	0.190 (+0.179)	0.110 (+0.095)	0.150 (+0.150)	0.114 (-0.041)	**0.140**	0.296	not eligible: valid rate
K3	interface: FFSS at TOA	0%	– (–)	– (–)	– (–)	– (–)	–	–	not feasible: all-NaN output
K4	interface: FFSS after AC (BOA)	93%	0.229 (+0.229)	0.140 (+0.140)	0.222 (+0.222)	0.079 (+0.015)	**0.168**	0.302	**eligible**

Each cell reads median |log10 error| (median bias) against AERONET-OC Level 2.0.

Paired against K0 (matchup-block bootstrap):

Candidate vs K0	Matchups	Red (B04)	Red difference (95% CI)	B02–B05	B02–B05 difference (95% CI)
K1	40	0.152 vs 0.247	-0.095 (-0.138 to -0.037)	0.148 vs 0.209	-0.061 (-0.089 to -0.038)
K2	41	0.150 vs 0.251	-0.101 (-0.152 to -0.046)	0.140 vs 0.213	-0.073 (-0.108 to -0.042)
K4	43	0.222 vs 0.251	-0.029 (-0.051 to -0.018)	0.168 vs 0.209	-0.041 (-0.052 to -0.021)

BEST, by the frozen rule, is K4: ACOLITE with its FFSS sky-dome interface correction applied after the atmospheric correction. It is the only candidate that meets every condition:

- **Red error.** 0.222 against 0.251 for K0 on the same matchups, and the CI excludes 0.
- **All bands (B02–B05).** 0.168 against 0.209.
- **Valid rate.** 93%, above the 91% floor.
- **No negative water reflectance.**
- **Inland turbidity guard.** 0.302, within K0's 0.279 + 0.03.

K1 and K2 would have fixed far more of the red band, but they lose retrievals.

- **The gain.** They cut the red error to 0.15 and the red bias from +0.25 to +0.15. Their B02–B05 error of 0.14–0.15 is the lowest of any candidate.
- **The cost.** Their valid rate is 87% and 89% on AERONET-OC, against a floor of 91%. They also keep only 40–42 of the guard stations, against K0's 52.
- **The likely reason:** the sky-reflection term they subtract pushes clear, dark water below zero, where ACOLITE masks it. This is inferred from where the losses fall; the flags were not broken down.
- **So the pre-registered valid-rate rule rules them out.**

K3 (FFSS at top of atmosphere) returns NaN on every scene. That is a failure inside ACOLITE: not feasible.

K0 reproduces Stage 11 exactly on the 3 check scenes.

RED-BAND RESIDUAL FIXED: NO.

- K4's red bias is +0.222 against K0's +0.254, a 13% cut; the bar for PARTIAL was 30%.
- **What the experiment does show:** subtracting sky reflection explains roughly 40% of the red over-brightness (K1/K2). The rest is presumably aerosol or adjacency error, which no interface option addresses.

4. Part 2: harmonising Sentinel-2C

Coefficients (all 12 pairs; S2C → S2A-equivalent; k is the B05 band-shape term on B06 – B04):

Band	TOA: gain, offset (ACOLITE path)	Applied to operational TOA	BOA: gain, offset (Sen2Cor path)
B02	0.994, -0.0026	no	0.992, +0.0007
B04	0.983, +0.0003	no	0.993, +0.0003
B05	0.979, +0.0002; k -0.059	yes	1.000, +0.0007; k -0.045
B08	0.987, -0.0005	yes	0.992, -0.0005
B8A	0.982, -0.0008	yes	0.992, -0.0008
B11	0.832, -0.0006	yes	0.829, -0.0010
B12	0.827, -0.0002	yes	0.847, -0.0005

P2 route flips on identical water, held-out pairs:

Tandem pair (held out)	Sen2Cor path: raw → harmonised	ACOLITE raw	ACOLITE V-full	ACOLITE V-op	ACOLITE V-pw
Chapala (13QFC)	3% → 0%	4%	3%	3%	3%
Chapala (13QGC)	7% → 1%	5%	3%	3%	2%
Pontchartrain (15RYP)	19% → 11%	0%	0%	0%	0%
Pontchartrain (16RBU)	56% → 15%	0%	0%	0%	0%
Mobile Bay (16RCU)	40% → 14%	1%	0%	0%	0%
Mobile Bay (16RDU)	26% → 18%	4%	0%	1%	0%
Tampa Bay (17RLL)	6% → 4%	75%	19%	62%	11%
Volta (30NZP)	12% → 8%	–	–	–	–
Kariba (35KNB)	0% → 0%	29%	1%	2%	4%
Kariba (35KPB)	0% → 0%	90%	3%	6%	9%
Nasser (36QUJ)	100% → 13%	–	–	–	–
Van (38SLH)	21% → 11%	14%	10%	10%	10%
Mean over pairs	**24% → 8%**	**22%**	**4%**	**9%**	**4%**

Residual band differences (median over held-out pairs):

Band	Sen2Cor raw C/A (C – A)	Sen2Cor harmonised	ACOLITE raw	ACOLITE V-pw
B02	0.989 (-0.0006)	0.992 (-0.0004)	0.978 (-0.0010)	1.000 (-0.0000)
B04	1.001 (+0.0001)	1.003 (+0.0001)	0.950 (-0.0005)	0.995 (-0.0003)
B05	0.953 (-0.0014)	1.010 (+0.0003)	0.873 (-0.0030)	1.001 (-0.0000)
B8A	1.076 (+0.0009)	1.001 (-0.0000)	0.768 (-0.0010)	1.015 (+0.0001)
B11	1.458 (+0.0021)	0.993 (-0.0000)	0.428 (-0.0001)	0.174 (-0.0006)
B12	1.276 (+0.0012)	0.981 (-0.0001)	0.321 (-0.0007)	0.313 (-0.0005)

Sen2Cor production path: the bands are fixed, but P2's decisions still flip.

- **The bands.** Harmonisation brings every band's held-out median within 2%: B05 1.010, B8A 1.001, B11 0.993, B12 0.981.

- **The decisions.** P2 still flips on **8.1%** of identical-water pixels, down from 24.1%. The flips concentrate in turbid pairs (Pontchartrain, Mobile Bay 11–18%).
- **Most likely why:** P2's residual test reacts to the pixel-level scatter between two sensors, which harmonising band medians cannot remove.

ACOLITE path: the adopted method (V-ρw) brings flips under the bar.

- **Flips.** P2's decisions on ACOLITE reflectance flip on **3.9%** of held-out pixels, down from 22.3% raw.
- **Residuals.** B02–B05 and B8A are within 1.5%.
- **The TOA method (V-op) is worse** at 8.8% (Tampa Bay 62%). It aligns the band medians, but not the dark pixels that ACOLITE's aerosol step reads.
- **The residual that misses:** B11, at −0.0006 against a ±0.0005 bar. ACOLITE sets water B11 to about zero, so the band carries little for water.
- **Coverage.** Two pairs, Lake Volta and Lake Nasser, have no water pixel valid in every run, so 10 of 12 pairs are scored.

S2C HARMONIZED: PARTIAL, by the frozen rule. Only one path is under 5%, and one ACOLITE residual (B11) misses its bar.

The caveat that matters.

- **Only the commissioning tandem is public.** V-ρw raises S2C's visible water reflectance by about 0.002, because commissioning S2C read darker through ACOLITE. Operational S2C may not need that.
- **Where it gets tested.** The corpus platform test (section 6) checks it on 2025 S2C.

5. Part 3: the corpus through the new reflectance

Retrieval rate (R1).

	ACOLITE valid 60 m disk	Sen2Cor production
All corpus rows (not Guntersville, 1,582)	**76%**	100% (by construction)
Clear	77%	100%
Normal	75%	100%
Stained	76%	100%
Limited	76%	100%

- **ACOLITE loses a quarter of the stations, evenly across classes.** Of the 379 stations without a valid disk:
  - **289** have every disk pixel flagged by ACOLITE's water mask;
  - **64** keep fewer than 5 valid pixels;
  - **24** crops produced no output, because the aerosol fit failed;
  - **2** scenes have no L1C.
- **Which flag.** The flag type was not recorded per station. At the one station inspected it was the negative-reflectance flag.
- **Sen2Cor production answers everywhere,** because every corpus row was selected on a valid Sen2Cor read.
- **Unknown in Part 5 is these stations.**

Turbidity from each path's red band (R2), as median |e_T| (bias), with the paired difference:

Rows	n (lakes)	ACOLITE median |e_T| (bias)	Sen2Cor median |e_T| (bias)	ACOLITE – Sen2Cor (95% CI)
All, same-day pass	200 (158)	0.258 (-0.116)	0.245 (+0.083)	+0.013 (-0.033 to +0.062)
All, pass within ±1 day	524 (337)	0.251 (-0.134)	0.210 (+0.042)	+0.041 (+0.004 to +0.076)
Secchi class: clear	66 (55)	0.259 (+0.183)	0.391 (+0.376)	-0.132 (-0.207 to -0.074)
Secchi class: normal	59 (51)	0.190 (-0.139)	0.174 (+0.026)	+0.015 (-0.086 to +0.093)
Secchi class: stained	35 (33)	0.235 (-0.208)	0.134 (-0.002)	+0.101 (+0.028 to +0.166)
Secchi class: limited	40 (35)	0.325 (-0.312)	0.230 (-0.177)	+0.095 (+0.028 to +0.217)
In-situ type: clear	108 (91)	0.224 (-0.022)	0.276 (+0.203)	-0.052 (-0.130 to -0.005)
In-situ type: algal	20 (18)	0.293 (-0.279)	0.175 (-0.082)	+0.118 (-0.020 to +0.224)
In-situ type: sediment	18 (16)	0.256 (-0.229)	0.148 (-0.028)	+0.108
In-situ type: mixed	27 (25)	0.332 (-0.330)	0.204 (-0.181)	+0.128 (+0.045 to +0.201)

- **Reflectance improved: NO by the frozen rule.** On same-day rows ACOLITE is +0.013 (CI includes 0); within ±1 day it is worse, +0.041 (CI excludes 0).
- **The pattern matters more than the total:**
  - **Clear water:** ACOLITE removes most of Sen2Cor's over-read (bias +0.18 against +0.38; error 0.26 against 0.39).
  - **Stained, limited, algal and mixed water:** ACOLITE reads turbidity lower than Sen2Cor (bias –0.21 to –0.33).
  - **Why this matters:** Sen2Cor's extra brightness was partly cancelling the turbid-water under-read that every processor shares.
- **Across the corpus ACOLITE's water reflectance is 17–33% darker than Sen2Cor's.** Blue is ×0.67, green ×0.83, red ×0.70 and 704 nm ×0.76. That fits Sen2Cor's AERONET-OC over-read.

The Part A split (R3), through the frozen champion:

Same-day rows with turbidity	ACOLITE FNU	Sen2Cor FNU
Radiometric share of Var(e_S)	60%	65%
Conversion share	40%	35%
Frozen champion median |e_S|	0.219	0.177

The frozen champion was fitted on Sen2Cor, so on ACOLITE it is worse (0.219 against 0.177). That is why Part 4 refits every empirical form.

By platform (R4):

Platform (turbidity rows ±1 day)	n	ACOLITE median e_T	Sen2Cor median e_T
sentinel-2b	307	-0.120	+0.032
sentinel-2a	204	-0.139	+0.062
sentinel-2c	13	-0.259	-0.169

6. Part 4: the simplest clarity engine

All candidates, held out by whole lakes (1,206 rows at 792 lakes with a valid ACOLITE disk, Guntersville excluded):

Candidate (on ACOLITE ρw)	Coverage	Median |log| (factor)	Median signed ft	Class accuracy	Eligible
1 champion, frozen	100%	0.190 (×1.55)	+0.66	0.56	yes
2 P1, frozen	76%	0.185 (×1.53)	-1.34	0.47	no (coverage)
3 champion form, refitted	100%	0.186 (×1.53)	-0.01	0.54	yes
4 N1 (red)	100%	0.186 (×1.53)	+0.02	0.54	yes
5 N2 (red + green/red)	100%	0.178 (×1.51)	+0.07	0.56	yes
6 C3 form, refitted	100%	**0.169 (×1.47)**	-0.03	0.60	yes
7 C6 form, refitted	100%	0.170 (×1.48)	+0.08	0.60	yes
8 P2, frozen	75%	0.169 (×1.48)	-0.92	0.49	no (coverage)
9 routed (nested)	99%	0.168 (×1.47)	+0.22	0.60	router
Production today: champion on Sen2Cor, same rows	100%	*0.219 (×1.66)*	*-0.75*	*0.46*	reference

Equivalence to the best, C3 (the lake-bootstrap CI's upper bound must be ≤ +0.010):

Simpler than the best (C3)	Difference in median |log|	95% CI	Equivalent (upper ≤ +0.010)
1 champion, frozen	+0.0216	+0.0060 to +0.0399	no
3 champion form, refitted	+0.0169	+0.0070 to +0.0304	no
4 N1 (red)	+0.0172	+0.0053 to +0.0285	no
5 N2 (red + green/red)	+0.0095	-0.0025 to +0.0238	no

Routing, judged against the engine:

	Routed	Engine (C3)
Median |log|, same rows	0.1677	0.1691
Relative gain	+0.8% (bar ≥ 10%)	
Difference 95% CI	-0.0108 to +0.0070	
Coverage	99%	100%

The engine is C3, refitted on ACOLITE reflectance: one model, no routing.

- **It is the best model,** and no simpler model is statistically equivalent. N2 comes closest, at +0.0095 with an upper bound of +0.024.
- **It passes the C3 guard in algal water:** 0.189, against 0.184 for N2, the best other model there (the bar is +0.01). So C3 is not an unconditional algal fallback.
- **Routing gains 0.8%** (CI includes 0), well short of the 10% bar, so the improved input does not need a router.
- **P1 and P2 score well on the rows they answer,** but answer only 75–76% of them, below the 90% needed for a single engine.

Where the gain comes from (exploratory, not pre-registered):

Same 1,206 rows, same folds	C3 on Sen2Cor	C3 on ACOLITE	Difference (95% CI)
Median |log|	0.200	**0.169**	-0.031 (-0.045 to -0.016)

On identical rows and folds, the reflectance change alone lowers C3's error by 0.031 log (16%).

Per class, held out, against production on the same rows:

True class / type	Rows (lakes)	ACOLITE C3 engine: error; signed; class accuracy	Production today, same rows
Clear ≥ 6 ft	574 (412)	**×1.56** (0.192); -3.3 ft; 0.75	×1.97 (0.295); -5.2 ft; 0.47
Normal 3–6 ft	286 (208)	**×1.27** (0.104); +0.3 ft; 0.64	×1.31 (0.119); +0.2 ft; 0.67
Stained 2–3 ft	173 (151)	**×1.38** (0.140); +0.9 ft; 0.29	×1.40 (0.147); +0.9 ft; 0.25
Limited < 2 ft	173 (137)	**×1.81** (0.257); +1.0 ft; 0.32	×1.98 (0.297); +1.3 ft; 0.29
In-situ algal	94 (77)	**×1.51** (0.178); +1.0 ft; 0.45	×1.57 (0.196); +1.2 ft; 0.43
In-situ sediment	64 (54)	**×1.37** (0.136); +0.6 ft; 0.48	×1.27 (0.104); -0.0 ft; 0.55
In-situ mixed	83 (61)	**×1.56** (0.194); +0.6 ft; 0.45	×1.62 (0.209); +0.9 ft; 0.33
Guntersville (reported apart; no selection)	42 (10)	×1.34 (0.128); -0.7 ft	×1.72 (0.236); -1.5 ft

- **Better** in clear water (×1.56 against ×1.97; the under-read shrinks from –5.2 to –3.3 ft), limited, algal and mixed water.
- **About the same** in normal and stained water.
- **Worse in sediment water** (×1.37 against ×1.27), consistent with ACOLITE's darker red.
- **Guntersville** (42 historical agency rows at 10 station groups, never used for selection): ×1.34 against ×1.72. This is historical only. Michael's survey remains the acceptance test.

Platform (condition b, held-out signed error of one platform minus the others):

Platform	Rows	Median signed log error minus the others'	95% CI
sentinel-2a	501	+0.016	-0.020 to +0.060
sentinel-2b	674	-0.009	-0.049 to +0.034
sentinel-2c	31	-0.103	-0.155 to +0.041

S2C reads 0.10 log low, but on 31 rows the CI includes 0, so condition (b) passes.

Condition (a) fails. On identical tandem water, the engine's class flips on 11.2% of pixels (bar 5%). The median change is only 0.16 ft:

Engine class decision on identical tandem water	Raw	V-full	V-op	V-pw (adopted)
Mean class-flip rate (10 pairs)	12.1%	8.5%	10.0%	**11.2%**
Median |Δft|	0.19	0.10	0.11	**0.16**

- **Harmonisation barely moves the class flips** (12.1% raw).
- **They most likely come from pixel-level scatter near the class boundaries** and near C3's NDCI = 0 branch switch, not from the S2C band bias, since harmonisation leaves them almost unchanged.
- **The tandem test is per pixel at 20 m,** while production reads a 60 m disk of about 28 pixels. So this flip rate probably overstates what a user would see. That was not tested, and the frozen rule still applies.

S2A/B/C PLATFORM-INVARIANT ENOUGH FOR CLARITY: NO.

7. Part 5: exact feet against clarity class

What a user would get, with exact feet shown only for classes the inner folds authorise:

What a user would get	Share of corpus rows
Exact feet (and the class)	15%
Class only (feet withheld)	61%
Unknown (no valid ACOLITE disk)	24%

- **Which classes get feet.** The inner folds authorised limited water in every fold and stained water in three. Normal was authorised once, and clear never; clear's predicted rows miss the ×1.41 and ±1.5 ft bars.
- **So most answers are a class, not a number:** 61% of rows get only a class, and 24% are Unknown.

The two historical capability scores (lake-held-out, Stage 10 bars; **not product readiness**):

Capability	Exact feet (published rows)	Feet verdict	Clarity class	Class verdict
Clear	37 published (5%); ×2.19; -4.4 ft	**NOT**	recall 0.58, precision 0.80, Unknown 23%	**PARTIAL**
Normal	47 published (12%); ×1.39; -0.8 ft	**NOT**	recall 0.48, precision 0.41, Unknown 25%	**NOT**
Stained	42 published (18%); ×1.17; -0.0 ft	**NOT**	recall 0.23, precision 0.32, Unknown 24%	**NOT**
Limited	113 published (49%); ×1.60; +0.8 ft	**NOT**	recall 0.24, precision 0.81, Unknown 24%	**NOT**
Algae	71 published (30%); ×1.53; +0.5 ft	**NOT**	class accuracy 0.33 vs constant 0.37	**NOT**
Sediment	33 published (36%); ×1.50; +0.5 ft	**NOT**	bright sediment called clear 0% (n 25); accuracy 0.34 vs 0.42	**PARTIAL**
Bottom-risk	–	**NOT**	no depth input	**NOT**
Score		**0.0 / 7 → 0/10**		**1.0 / 7 → 1/10**

- **EXACT-FEET CAPABILITY: 0/10.**
  - Feet are published too rarely: under 50% coverage in every class.
  - Where they are published for true clear water, it is because the model called it stained or limited. Those answers are badly wrong (–4.4 ft).
- **CLARITY-CLASS CAPABILITY: 1/10.**
  - **Clear is PARTIAL.** Precision is 0.80, but recall is 0.58 once the 23% Unknown counts as a miss. On retrieved rows alone it is 0.75.
  - **Sediment is PARTIAL:** no bright-sediment station is called clear.
  - **Stained and limited fail on recall** (0.23 and 0.24).

The same rubric on other inputs (exploratory; answered rows, and Unknown where noted):

Input and rows (exploratory)	Rows	Unknown	Median |log| (answered)	Class accuracy (answered)	Clear recall / precision	Normal	Stained	Limited
Production today (champion on Sen2Cor), all rows	1,582	0%	0.225	0.45	0.46 / 0.77	0.68 / 0.31	0.24 / 0.26	0.26 / 0.67
C3 on Sen2Cor, all rows	1,582	0%	0.198	0.52	0.69 / 0.75	0.57 / 0.32	0.19 / 0.25	0.17 / 0.78
C3 on Sen2Cor, the ACOLITE rows	1,206	0%	0.200	0.53	0.71 / 0.74	0.55 / 0.32	0.22 / 0.32	0.23 / 0.81
C3 on ACOLITE, the ACOLITE rows	1,206	0%	0.169	0.60	0.75 / 0.80	0.64 / 0.41	0.29 / 0.32	0.32 / 0.81
C3 on ACOLITE, all rows (Unknown where no disk)	1,582	24%	0.169	0.60	0.58 / 0.80	0.48 / 0.41	0.23 / 0.32	0.24 / 0.81

- **Coverage is one limit.** With a disk at every station, C3 on ACOLITE would clear the clear-class SUPPORTED bar (recall 0.75, precision 0.80).
- **Stained and limited water are the real capability gap.** They sit at recall 0.17–0.32 on every input, whether Sen2Cor or ACOLITE, champion or C3.

8. Part 6: the blind external certification set

The protocol was written and hashed before any site or visit file was fetched (`cert/CERT_PROTOCOL_STAGE12.md`, 15:04:45Z).

- **Sources:**
  - **US core:** EPA National Lakes Assessment 2022 and 2017, from EPA's own files. Neither is in the 2018–2026 WQP pull used for development.
  - **Cool-season stress test:** Finland's SYKE Vesla open data.
- **Eligibility and matching:**
  - lakes within 5 km of any development station are excluded, as is Guntersville;
  - NLA visits duplicated in the WQP pull are dropped;
  - each visit needs a production-gated Sentinel-2 pass within ±1 day;
  - at most 3 observations per lake;
  - every date is ≤ 2025-12-31, so the set is disjoint from P2's fresh holdout.
- **Blindness.**
  - Only site, visit and event-key tables were fetched.
  - No NLA Secchi file was downloaded.
  - SYKE queries never selected the value field.
  - `cert/blind_sample.csv` (sha256 `111cff44…`) lists IDs, dates and satellite items only.

Blind checks:

Blind check	Value	Bar	
Matched observations	289 (NLA 2017 52, NLA 2022 113, SYKE 124)	≥ 400	**fail**
Distinct lakes	272	≥ 100	pass
US states (US core)	39	≥ 20	pass
Largest state share (US core)	9.7%	≤ 15%	pass
October–April observations	4.2% (12 lakes)	≥ 15% across ≥ 20 lakes	**fail**
Expected clear (US core, planning prior)	77	≥ 40	pass
Expected normal (US core, planning prior)	41	≥ 40	pass
Expected stained (US core, planning prior)	24	≥ 60	**fail**
Expected algae (US core, planning prior)	36	≥ 30	pass
Expected sediment (US core, planning prior)	13	≥ 20	**fail**
Expected limited (US core, planning prior)	23	≥ 25	**fail**

BLIND EXTERNAL CERTIFICATION SET READY WITHOUT USER SURVEY: NO.

- **What passes:** geography and lake count. There are 272 lakes in 39 states plus Finland, with no state above 10%.
- **What fails:**
  - **Total:** 289 matched observations against the 400 bar.
  - **Match rate.** Only 11% of NLA visits have a clear pass within a day that survives the production gates, and 2017 is thin before Sentinel-2B.
  - **Cool season:** 4% (NLA samples only in summer, and Finnish October and November are cloudy).
  - **Expected stained, sediment and limited counts.**
- **What would complete it,** in order of independence:
  - SYKE extended from 400 lakes to all 3,705 with eligible events (cool season and humic stained water, but out of domain);
  - a request to TVA or USACE districts for their cool-season reservoir records;
  - a pre-registered, sealed tier of WQP records that were downloaded in Stage 1 but never analysed. That is your call, because those values already sit on this Mac.

9. The path to 9/10

Product readiness stays 2/10. Under the Stage 10 rule only fresh evidence moves a capability.

Step	The Stage 12 finding it rests on	What it would move
1. Recover the 24% of lost retrievals	ACOLITE gives a valid disk at 76% of stations. Its water mask flags whole disks at 289 stations and most of the disk at 64 more, and the aerosol fit fails on 24 crops	Clear to SUPPORTED on the class rubric. On retrieved rows its recall is already 0.75
2. Give the class decision an uncertainty band at the 3 and 6 ft boundaries, and re-test S2A/S2C on disk-averaged pixels	11% of pixels flip class on identical water, with a median change of only 0.16 ft	Platform invariance
3. Find a signal for stained and limited water	Recall 0.17–0.32 on every input and model; routing and refits do not help	Stained and limited, the capabilities that separate 4/10 from 9/10
4. Complete the blind set	289 observations; stained, sediment, limited and cool season short	Certification without anyone surveying
5. Certify once, on the blind set and on P2's fresh data	The Stage 10 bars, unchanged	Readiness

- **Step 1 options, to pre-register:** ACOLITE mask variants, or a hybrid. The hybrid uses ACOLITE where it reads the water, and C3 on Sen2Cor elsewhere, flagged as lower authority. C3 on Sen2Cor is ×1.58 against production's ×1.68 on every row.
- **What 9/10 needs:** clear, normal, stained, algae, sediment and limited all SUPPORTED on fresh data. Bottom-risk stays NOT until a depth layer exists (the Navionics track).

Files

Everything is in `scratchpad/stage12/`:

Path	Contents
`PREREGISTRATION_STAGE12.md`, `FREEZE_STAGE12.json`, `ADDENDUM_*.md`	The frozen plan, its hashes and the addenda
`exp1/`	The interface experiment: guard sample, runner, per-scene results, `score.json`
`harm/`	The harmonisation fit, leave-one-pair-out validation, ACOLITE tandem runs, `score_tandem.json`
`corpus/`	The corpus driver, elevations, per-scene ACOLITE samples
`analysis/`	`reflectance.py`, `models12.py`, `classes12.py`, tables and JSON
`cert/`	The certification protocol, candidate builders, matching, `blind_sample.csv` and its hash
`tools/`	The ACOLITE runner with overrides, and the L1R harmoniser

BEST WATER-REFLECTANCE CONFIGURATION: ACOLITE 20260421.0 (DSF, residual-glint correction, the FFSS sky-dome interface correction after AC; candidate K4) with surface pressure from each lake's USGS elevation, 20 m, and Sentinel-2C harmonised at the water-reflectance level (V-ρw)

RED-BAND RESIDUAL FIXED: NO

S2C HARMONIZED: PARTIAL

S2A/B/C PLATFORM-INVARIANT ENOUGH FOR CLARITY: NO

BEST CLARITY ENGINE AFTER CORRECTION: C3 refitted on ACOLITE reflectance (two red-band branches split on NDCI > 0), a single model with no routing: held-out ×1.47 (median |log| 0.169) against ×1.66 for today's production on the same rows, but only at the 76% of stations where ACOLITE retrieves water

CLEAR WATER ERROR: ×1.56 (median |log| 0.192), median −3.3 ft, class accuracy 0.75 (n 574; production ×1.97, −5.2 ft)

NORMAL WATER ERROR: ×1.27 (0.104), median +0.3 ft, class accuracy 0.64 (n 286; production ×1.31, +0.2 ft)

STAINED WATER ERROR: ×1.38 (0.140), median +0.9 ft, class accuracy 0.29 (n 173; production ×1.40, +0.9 ft)

LIMITED WATER ERROR: ×1.81 (0.257), median +1.0 ft, class accuracy 0.32 (n 173; production ×1.98, +1.3 ft)

ALGAL WATER ERROR: ×1.51 (0.178), median +1.0 ft (n 94, in-situ algal; production ×1.57, +1.2 ft)

SEDIMENT WATER ERROR: ×1.37 (0.136), median +0.6 ft (n 64, in-situ sediment; production ×1.27, −0.0 ft, so worse than production)

EXACT-FEET CAPABILITY: 0/10 (historical lake-held-out; feet are authorised for 15% of rows, and no class reaches 50% published coverage)

CLARITY-CLASS CAPABILITY: 1/10 (historical; clear and sediment PARTIAL; 24% Unknown; stained and limited recall 0.23–0.24)

BLIND EXTERNAL CERTIFICATION SET READY WITHOUT USER SURVEY: NO

CURRENT PATH TO 9/10: Readiness stays 2/10. First recover ACOLITE's 24% lost retrievals (pre-registered mask variants or a Sen2Cor-C3 fallback). Add a class-boundary uncertainty band so S2A/S2C scatter stops flipping classes. Find a signal for stained and limited water, the weakest classes on every input. Complete the blind set (≥ 400 observations, stained and cool-season water) from SYKE's full lake list plus agency records or a sealed never-analysed WQP tier. Then certify once, on that set and on P2's fresh data, against the Stage 10 bars.